

Computational Representations for Graphical User Interfaces

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ABSTRACT

Graphical User Interfaces (GUIs) have been widely used in daily life. To enhance GUI design and interaction experience on GUIs, it is important to understand GUIs and understand how individuals interact with them. Consequently, my thesis focuses on applying computational approaches to improve our understanding of GUIs and user interactions. First, I introduce novel GUI representations to capture the visual, spatial, and semantic factors of GUIs and improve the performance of downstream GUI tasks. Second, I simulate how users visually engage with GUIs to understand user interactions to help inform the design of GUI representations. Third, based on the understanding of GUIs and interactions on GUIs, I develop language language representations aimed at assisting users in understanding and more effectively interacting with GUIs.

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1 INTRODUCTION

Graphical User Interfaces (GUIs) have played an important role in enhancing digital interactions. Understanding GUIs and how users interact with them is essential for improving GUI design and user experience on GUIs [18–20]. In my thesis, I employ computational approaches to understand GUIs and user interactions with them.

Different graphical user interface (GUI) representations can capture various aspects of GUI design. Deep learning approaches heavily depend on how data are represented, and different representations can capture the different factors of GUI design. The choice of data representation for GUIs can affect the capabilities of downstream tasks within graphical user interfaces. The effectiveness of deep learning-based methods often depends on the choice of data representation. Present deep learning-based GUI methods often require the design of preprocessing pipelines and data transformations, which can be time-consuming and may struggle to extract discriminative information from the data. A well-designed representation can leverage prior knowledge to address this limitation.

Many downstream tasks on GUIs require the use of suitable representation. Recent computational approaches improved the GUI design workflows by giving suggestions on GUIs [14, 25, 27, 38, 44],

adapted GUIs to different devices and user preferences, or user actions [10, 11, 13, 15, 28, 32, 33, 42, 43], reverse engineered GUIs to understand UIs and improve accessibility [9, 22, 35, 45, 51]. They use various GUI representations to solve individual downstream tasks but still lack a unified representation for various tasks.

Existing GUI representations have the limitation that they cannot capture all the visual, spatial, and semantic factors of GUIs. Some representations focus on textural content [29, 30], while others pay attention to visual appearance [2, 8, 37]. They only emphasize certain aspects of GUI properties while neglecting others. I aim to address it by bridging the gap between textual content, visual appearance, and layout-based constraints to enhance the interpretability and performance of downstream GUI tasks. The second aspect of my research focuses on understanding and simulating user interactions with GUIs to gain insights into how users visually engage with these interfaces. I aim to use eye tracking prediction to inform the design of representations by understanding how GUIs are perceived. Previous work often focused on eye tracking proxies, such as webcam or mouse, or concentrated on specific design types like mobile GUIs. Instead, I focus on simulating real eye tracking using the data collected from eye trackers and covering various GUI categories. Finally, building on our understanding of GUIs and user interactions, I developed a language representation with the goal of assisting users in better understanding and interacting with GUIs, serving as a UI-focused instruction-following visual agent.

Thus, the main research objectives of my thesis are

- To develop efficient GUI representations that capture visual, spatial, and semantic structures of GUIs for downstream tasks.
- To explore the simulation of human behaviors, such as eye tracking, for gaining insights into user interactions with GUIs, reducing the dependency on human analysis and user studies.
- To create a language agent that interactively assists users in understanding and interacting with GUIs.

2 RESULTS AND CONTRIBUTIONS

2.1 GUI Representation

Present-day graphical user interfaces (GUIs) consist of various arrangements of text, images, and interactive components. The challenge of effectively conveying a GUI's visual, spatial, and semantic structure in computational design remains unsolved. Current GUI representations have limitations as they tend to focus on specific aspects of GUI properties while overlooking others. Some previous methods focused on capturing textual content within GUIs [29, 30]. However, these approaches neglected the visual appearance and

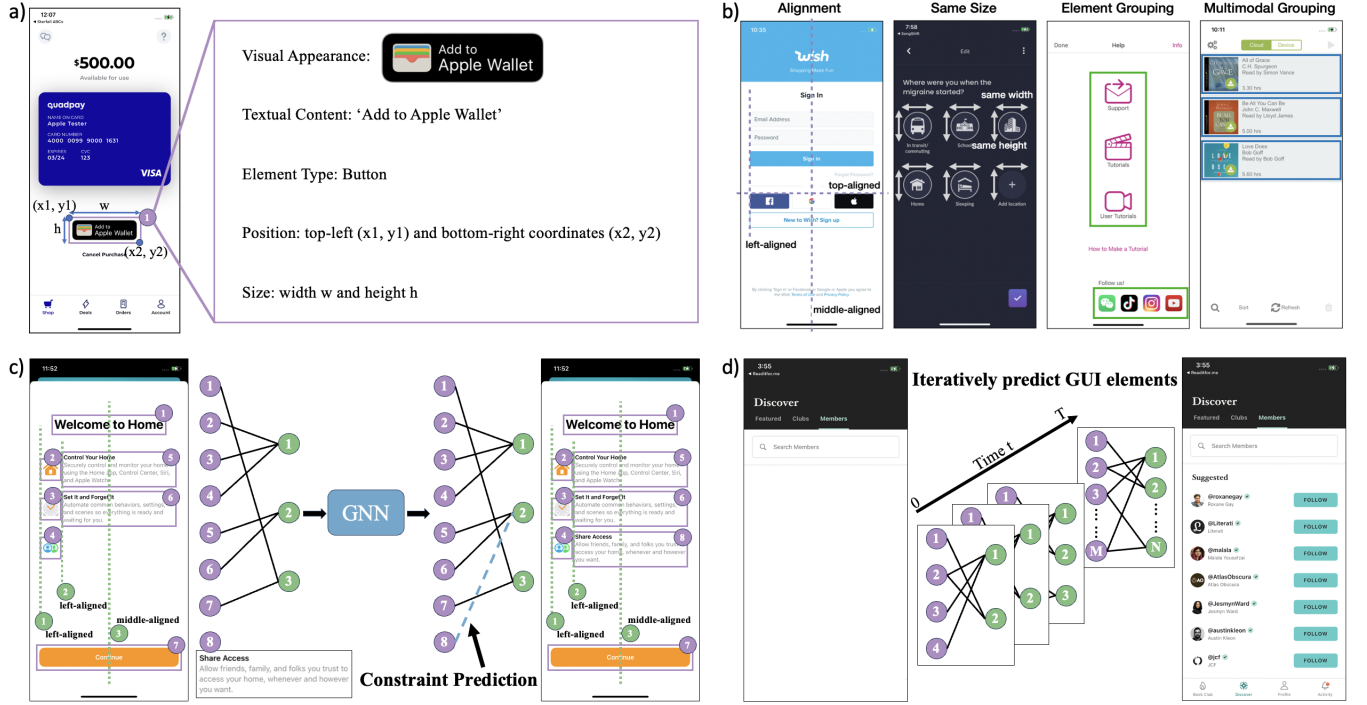


Figure 1: Our graph-based GUI representation connects between GUI element properties (a) and constraints (b) to capture the visual-spatial-semantic structure of a GUI. It is a bipartite graph containing element nodes (colored purple) representing the GUI elements' properties and constraint nodes (colored green) that can be fed into graph neural networks (GNNs) (c). We show that it can help GUI design by iteratively suggesting element sizes and positions (d).

types of GUI elements. On the other hand, some methods prioritized visual appearance and the types of GUI elements [2, 8, 37] but often neglected textual content. Consequently, they may treat GUIs with similar structural and visual features differently due to variations in textual content. Some attempts to bridge this gap between textual content and visual appearance [31, 34] require substantial task-specific datasets and manual data labeling. In contrast to neural networks, constraint-based approaches have also been explored, allowing explicit rules for element and layout constraints [15, 22, 23, 41]. While constraint-based approaches can provide explicit representations of GUIs and enhance the interpretability of the representation, they still often require the labor-intensive process of manually constructing constraints.

To bridge the gap in GUI representation, we proposed Graph4GUI [24], a bipartite graph-based approach that integrates the properties of GUI elements with their layout-based constraints (Figure 1). This bipartite graph structure expresses each GUI element and its connections to others through element and constraint nodes. Our approach, Graph4GUI, offers a solution that considers not only textual content, visual appearance, and GUI element types but also the constraints and interrelationships between GUI elements. The incorporation of graph neural networks (GNNs) facilitates the learning of domain-specific representations from this graph-structured data, as it aggregates information from neighboring nodes, facilitating the exchange of knowledge between element and constraint nodes. This approach recognizes that individual elements interact with

the overall layout, such as icons often grouped and aligned within toolbars. The utilization of GNNs enables us to capture both the GUI layout's structure and the unique properties of GUI elements. To assess the effectiveness of this graph-based representation, we applied it to an autocompletion task, enhancing efficiency by iteratively predicting the locations of unplaced GUI elements and providing confidence levels to assist designers in prioritizing element placement. Compared to prior approaches, our graph-based representation can better understand GUI structures, enhancing the model's interpretability and explainability.

2.2 Simulating User Interactions for GUI Representation

The most distinctive aspect of GUIs compared to natural images is that GUIs are often interactive. Thus, it is important to understand how users interact with GUIs, such as how users look at GUIs. Thus, to better inform the design of GUI representation, we also need to understand and simulate how users interact with GUIs. Eye tracking data can be used to generate heatmaps and fixation maps, which offer visual representations of how users visually engage with and navigate through GUIs, determining where users focus their visual attention when interacting with a GUI. This helps GUI designers understand which elements of the GUI draw the most attention and which ones are often overlooked. Designers can then optimize the placement of important features or content to align with user attention.

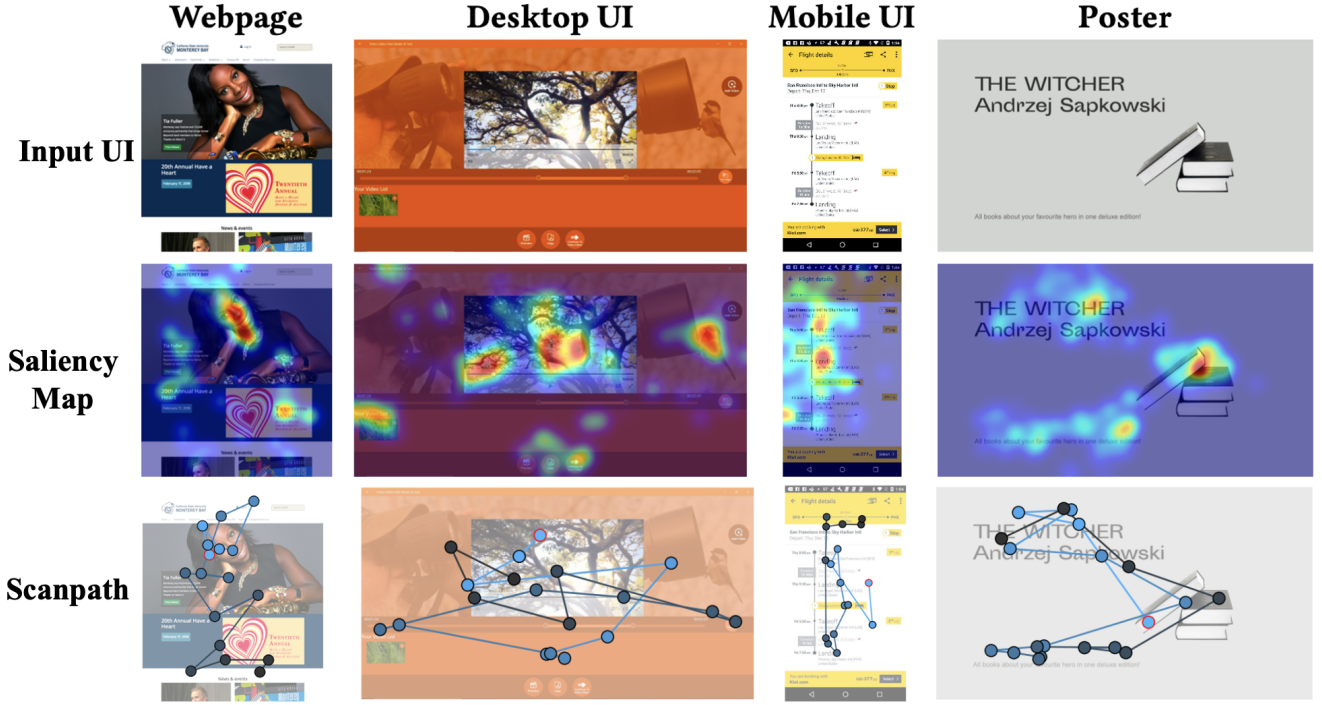


Figure 2: We developed models predicting saliency maps and scanpaths on GUIs.

To understand this, we collected and analyzed a novel eye-tracking dataset, UEyes [16, 17], using a high-fidelity in-lab eye tracker. This dataset includes multi-duration saliency maps and scanpaths generated from the eye movements of 62 participants who interacted with a diverse set of 1,980 user interfaces. Compared to previous eye-tracking datasets, which were often limited in size [3, 50] and often concentrated on specific design types like mobile GUIs [26], our UEyes dataset comprises high-quality eye-tracking data covering various GUI categories, including webpages, mobile interfaces, desktop applications, and posters. We further analyzed saliency-related patterns across the GUI types and created improved models for predicting saliency maps or scanpaths, simulating how users perceive GUIs. These predictive models predict saliency maps and scanpaths, respectively (Figure 2). They offer insights for GUI designers by estimating where users are likely to direct their gaze within a GUI and enable designers to update their GUIs to emphasize important areas. With visual saliency models, designers can improve their designs by making decisions based on how users are likely to perceive their GUIs [4]. Designers can use these visual saliency models to make decisions based on how users are likely to perceive their GUIs. Additionally, the predictive models for scanpaths, which capture the sequence and temporal dynamics of fixations, assist designers in understanding and modifying the visual flow within their designs, encouraging users to engage with GUI elements in the desired sequence.

Furthermore, individuals exhibit diverse gaze patterns influenced by factors such as prior exposure and learning approaches. Thus, we further proposed a novel deep Reinforcement Learning (RL) model with a Transformer architecture, which advances the predictive

modeling of personalized scanpaths in GUIs. It predicts both spatial and temporal scanpath characteristics, capturing users' viewing behaviors and long-range dependencies. It predicts a sequence of fixation points for a given GUI image, outperforming previous models. Additionally, it generates personalized scanpaths from a few user-specific scanpath samples, reflecting individual preferences. Unlike methods relying on mouse movements or manual annotations, our model replicates actual eye tracker-recorded scanpaths.

2.3 Language Representation for GUIs

Recent advancements in large language models (LLMs) and vision-language models (VLMs) have opened up new opportunities for computational understanding and interactions with GUIs [47–49]. As natural language is one of the main mediums for human communication and interaction, we can construct language representation to enhance the user experience of understanding and interacting with GUIs. Using text descriptions of GUIs alone with LLMs leaves out the rich visual information of the GUI. Although LLMs have demonstrated remarkable abilities to comprehend task instructions in natural language in many domains [6, 39, 40, 46, 52], fusing visual with textual information is important to understanding GUIs as it mirrors how many humans engage with the world. One approach that has sought to bridge this gap when applied to natural images is Vision-Language Models (VLMs), which accept *multimodal* inputs of both images and text, typically output only text, and allow for general-purpose question answering, visual reasoning, scene descriptions, and conversations with image inputs [1, 5, 7, 36]. However, the performance of these models on GUI tasks falls short compared to natural images because of the lack of GUI examples

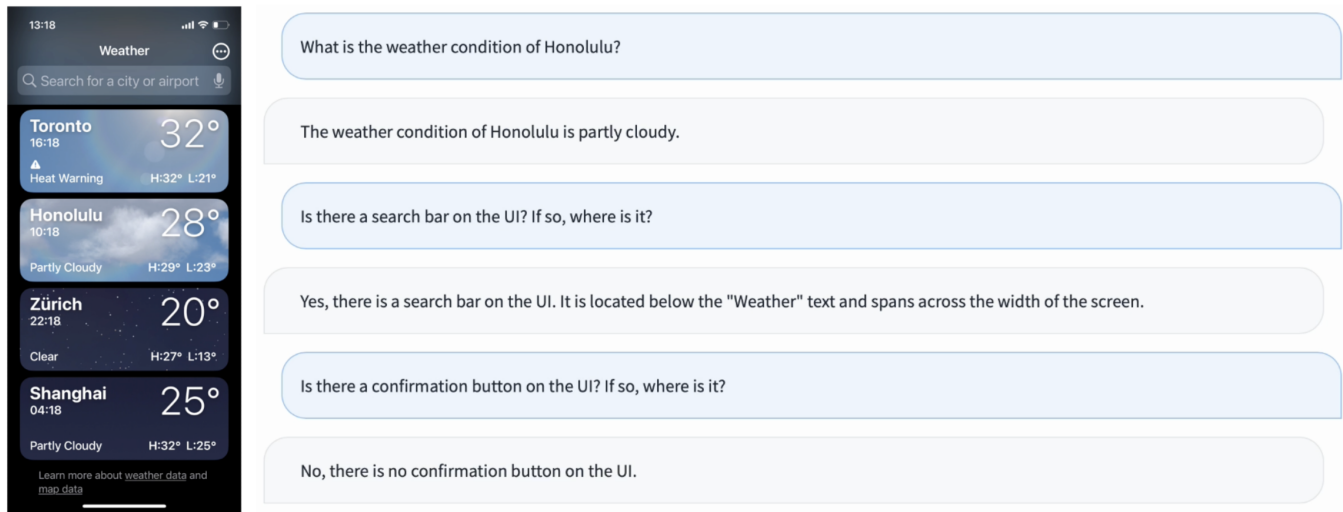


Figure 3: We created a language agent for GUIs, which can perform many UI-related tasks, including conversations, detailed descriptions, listing available actions, predicting UI action outcomes, selecting UI elements, and goal-based planning.

in their training data. As shown in Figure 3, we generated paired text-image data to train a VLM model, ILuvUI [21], which is a GUI-focused instruction-following visual agent. It can perform many GUI-related tasks, including conversations, detailed descriptions, listing available actions, predicting GUI action outcomes, selecting GUI elements, and goal-based planning.

3 FUTURE DIRECTIONS

3.1 GUI Representation

I plan to focus on extending the graph-based GUI representation to apply to more diverse GUI-related tasks. I have demonstrated the effectiveness of our graph-based GUI representation in the context of GUI autocompletion; my future work will explore its applicability in more diverse domains, such as evaluation and improvement of accessibility, as well as the detection of hierarchy and grouping within GUIs. For accessibility, I aim to represent accessibility needs as constraints [12] and use such constraints to train and predict layout constraints, potentially leading to improvements in accessibility. In addition, I also plan to employ the graph-based GUI representation to detect hierarchy and grouping within GUIs with high accuracy. I intend to make our method and model open-source and hope to open up future opportunities for GUI-related research.

3.2 Simulating User Interactions for GUI Representation

Our current model is only designed for free-viewing eye tracking within GUIs. My future research can extend to scope to solve task-based eye tracking problems, such as visual search tasks on GUIs. This extension will help us better understand human behaviors and interactions on GUIs, especially predict how users perform tasks while navigating through GUIs. Future work can also explore gaze patterns on GUIs with dynamic elements on GUI transitions and other types of user interactions on GUIs, such as text entry.

3.3 Language Representation for GUIs

I plan to improve both the quality of the language dataset on GUIs and the vision-language model to create a better language agent for GUIs. I will analyze our current dataset and create synthetic data based on the weaknesses, such as reducing hallucination and including more types of tasks for GUIs. On the other hand, I plan to improve the performance by building better vision-language models that can accept high-resolution GUIs. In addition, I also plan to support machine-interpretable output, such as JSON. For example, applications like UI navigation and software testing can benefit from machine-interpretable output.

4 DISSERTATION STATUS AND LONG TERM GOALS

I am currently a PhD student at Aalto University and the Finnish AI Center (FCAI) in Finland under the supervision of Professor Antti Oulasvirta (primary) and Professor Vikas Garg. I aim to have my Ph.D. defense in the spring of 2025. I have never attended any other doctoral consortium. My long-term goal is to get a tenure-track faculty job or be a research scientist in a relevant industrial group.

5 BENEFITS AND CONTRIBUTION STATEMENT

I hope to gain valuable feedback for my dissertation direction from the mentors and peers at the CHI2024 Doctoral Consortium. I think it is an opportunity to engage in discussion with peers to refine my research directions. In addition, I will be honored to attend the CHI2024 Doctoral Consortium to connect with other Ph.D. students working on different topics of HCI to expand my network within the academic community and potentially explore future collaborations. I will actively engage in discussions, provide feedback to other attendees, and communicate with them. Together with other peers, I will help create a supportive community and keep the connection after the Doctoral Consortium.

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